

2.7 The derivative as a function

Let $f(x)$ be a function such that given any number x ,

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (1)$$

exists.

We consider the function $f'(x)$ defined by

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (2)$$

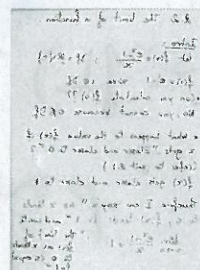
and called the derivative of f .

Remark: The domain of f' is all $x \in \mathbb{R}$ such that (1) exists.

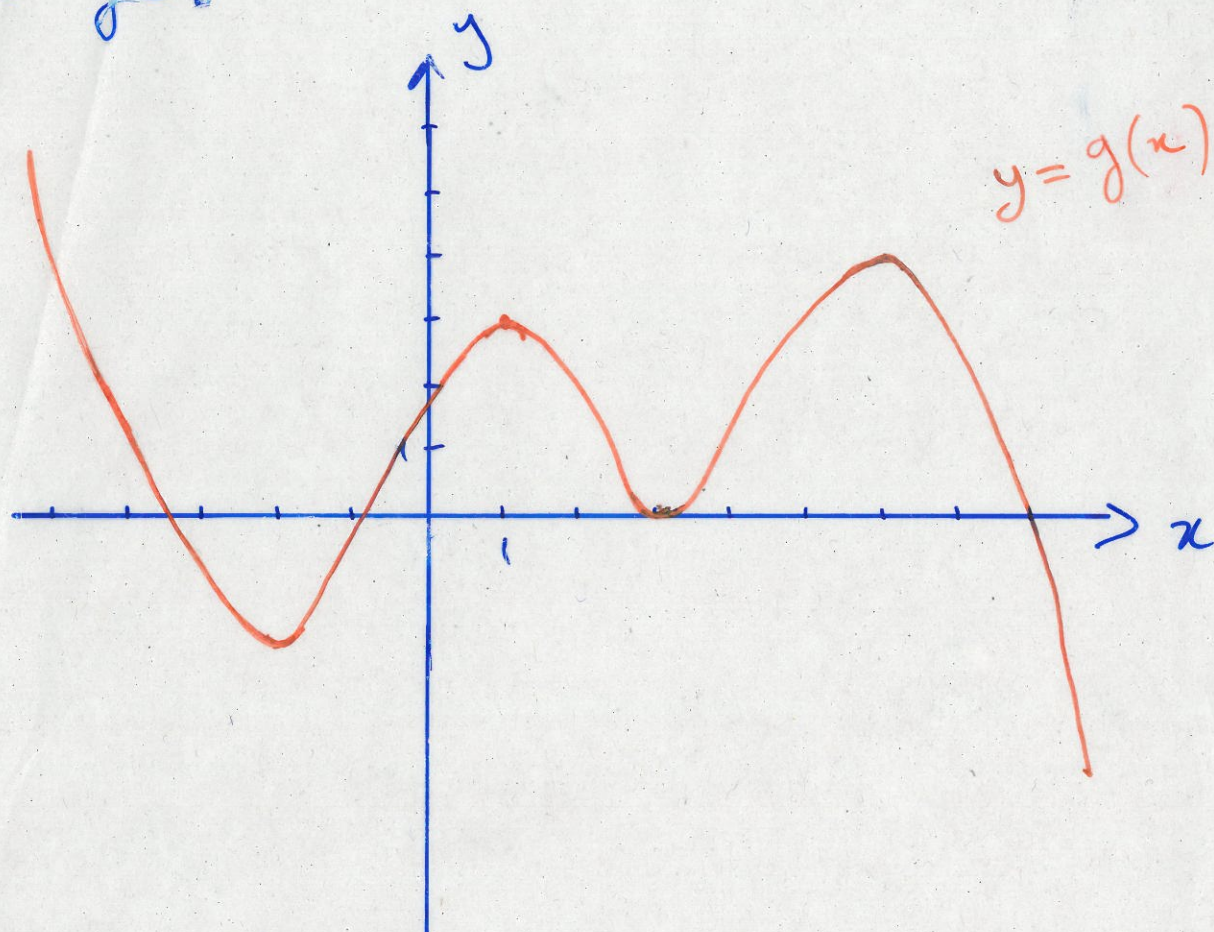
Example 1: Determine $f'(x)$ using (2), given f below, and determine the domain of f and f' .

(a) $f(x) = \sqrt{x}$

(b) $f(x) = \frac{1}{x}$



Example 2 : The graph of g is given below.
Use it to sketch the graph of its derivative g' .



(Hint : use the sign of the slope of the tangent line to $y = g(x)$ at every point.)

Notations : If $y = f(x)$ then

$$(3) f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx} f(x) = Df(x) = D_x f(x)$$

D and $\frac{d}{dx}$ are called differentiation operators.

Example 1

$$(a) \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \quad \left(\begin{array}{c} 0 \\ 0 \end{array} \right)$$

$$\frac{\sqrt{x+h} - \sqrt{x}}{h} = \frac{\sqrt{x+h} - \sqrt{x}}{h} \times \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}}$$

$$= \frac{x+h - x}{h(\sqrt{x+h} + \sqrt{x})}$$

$$= \frac{h}{h(\sqrt{x+h} + \sqrt{x})}$$

$$= \frac{1}{\sqrt{x+h} + \sqrt{x}}$$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} = \frac{1}{2\sqrt{x}}$$

$$\Rightarrow f'(x) = \frac{1}{2\sqrt{x}}$$

$$Df = [0, +\infty) \quad , \quad D_{f'} = (0, +\infty)$$

$$(b) D_f = \mathbb{R} \setminus \{0\}$$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{1}{x+h} - \frac{1}{x} \right)$$

$$\frac{1}{h} \left(\frac{1}{x+h} - \frac{1}{x} \right) = \frac{1}{h} \left(\frac{x - (x+h)}{x(x+h)} \right)$$

$$= \frac{-h}{hx(x+h)} = -\frac{1}{x(x+h)}$$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{-1}{x(x+h)} = -\frac{1}{x^2}$$

$$f'(x) = -\frac{1}{x^2} \quad \text{and} \quad D_{f'} = \mathbb{R} \setminus \{0\}$$

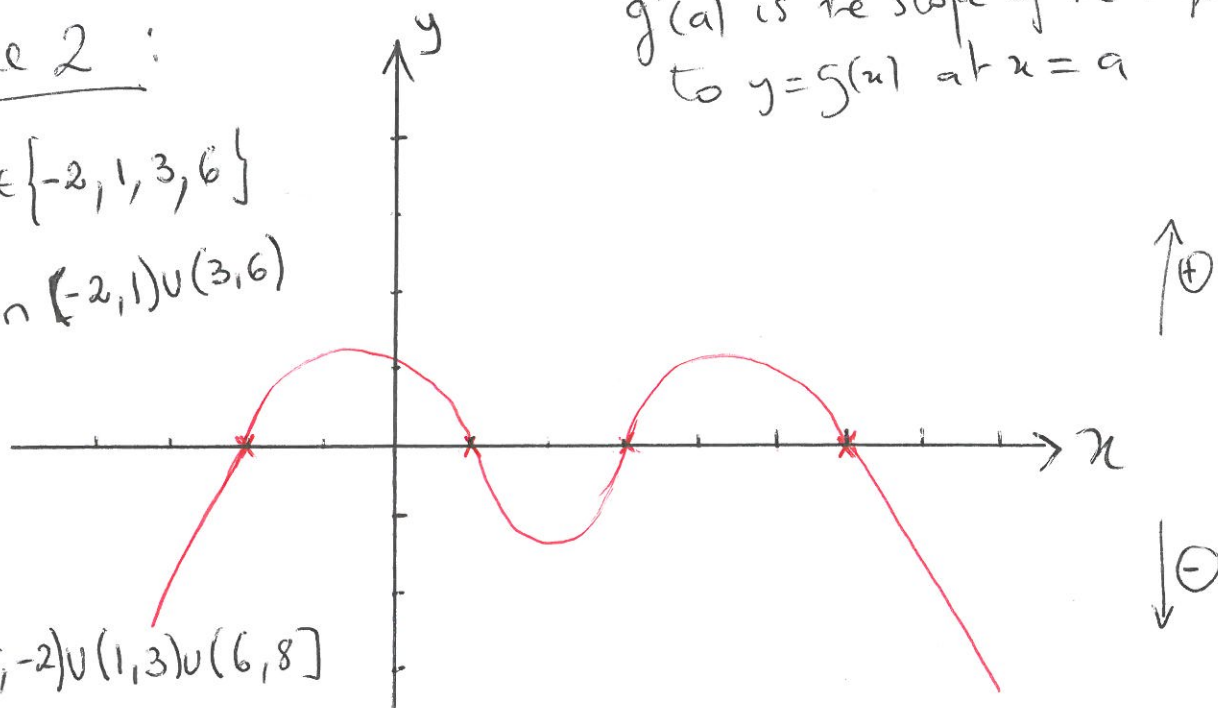
Example 2:

$$g'(a) = 0, a \in \{-2, 1, 3, 6\}$$

$$g'(a) > 0 \text{ on } (-2, 1) \cup (3, 6)$$

$$g'(a) < 0 \text{ on } [-5, -2) \cup (1, 3) \cup (6, 8]$$

$g'(a)$ is the slope of the tangent to $y = g(x)$ at $x = a$



Definition 2: A function is differentiable at a if $f'(a)$ exists. It is differentiable on a open interval (a, b) [or $(a, +\infty)$ or $(-\infty, a)$ or $(-\infty, \infty)$] if it is differentiable at every number in the interval.

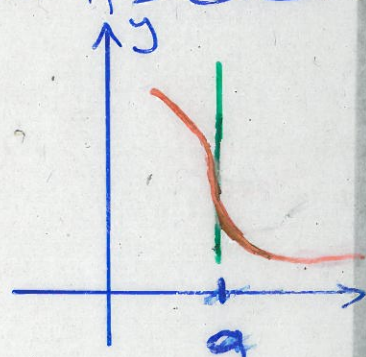
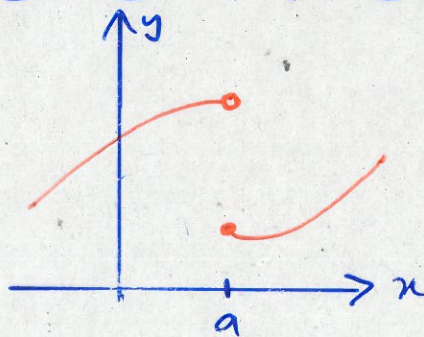
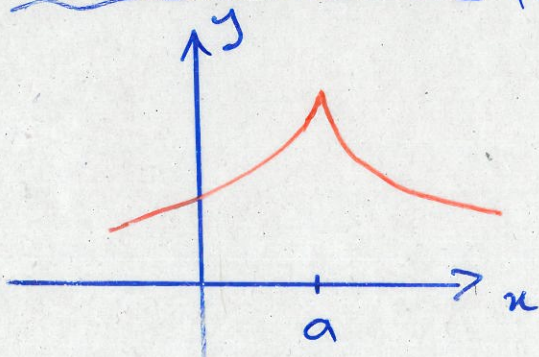
Geometric meaning:

f is differentiable at $x=a$ if you can draw a tangent line with a finite slope at $(a, f(a))$.

Theorem 1: If f is differentiable at a , then f is continuous at a .

"Proof": (see board)

How can a function fail to be differentiable?



Show that:

f differentiable at $a \Rightarrow f$ continuous at a

"Must be known for test purpose"

$$\text{(left)} \quad f \text{ differentiable} \Rightarrow f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \\ = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$\text{Moreover} \quad \lim_{x \rightarrow a} f(x) = f(a) \checkmark$$

$$\text{Hence} \quad \lim_{x \rightarrow a} [f(x) - f(a)] = \lim_{x \rightarrow a} \left([f(x) - f(a)] \times \frac{x-a}{x-a} \right) \checkmark$$

$$= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \times (x - a)$$

$$= f'(a) \times 0$$

$$= 0 \checkmark$$

$$\lim_{x \rightarrow a} f(x) \checkmark = \lim_{x \rightarrow a} [f(x) - f(a) + f(a)]$$

$$= \lim_{x \rightarrow a} [f(x) - f(a)] + \lim_{x \rightarrow a} f(a)$$

$$= 0 + f(a) \checkmark$$

$$\lim_{x \rightarrow a} f(x) = f(a) \quad \text{so}$$

f is continuous at a . (right)

Steps:

(1) use definition of differentiability at a (with $\lim_{x \rightarrow a}$)

(2) show that $\lim_{x \rightarrow a} [f(x) - f(a)] = 0$

(3) work out $\lim_{x \rightarrow a} f(x)$ as $\lim_{x \rightarrow a} [f(x) - f(a) + f(a)]$

and show that is it $f(a)$.

Example 3: Where is the function not differentiable?

(a) $f(x) = |x-2|$

(b) $f(x) = \sqrt{x+3}$

(c) $f(x) = \begin{cases} x+1, & x < 1 \\ 4, & x \geq 1 \end{cases}$

Since $f'(x)$ is a function, the derivative of $f'(x)$ is also a function, that is the second order derivative of f , denoted

$$f''(x) = \frac{d}{dx} [f'(x)] = \frac{d^2}{dx^2} f(x) \quad (4)$$

The n^{th} -order derivative of f is denoted

$$f^{(n)}(x) = \frac{d}{dx} [f^{(n-1)}(x)] = \frac{d^n}{dx^n} [f(x)] \quad (5)$$

Example 3

$$(a) \frac{f(x+h) - f(x)}{h} = \frac{|x+h-2| - |x-2|}{h}$$

$$\left[\begin{array}{l} f'(x) = 1, \text{ if } x > 2 \\ f'(x) = -1, \text{ if } x < 2 \end{array} \right] = \begin{cases} \frac{x+h-2 - (x-2)}{h} & x > 2 \\ \frac{-x-h+2 - (-x+2)}{h} & x < 2 \end{cases}$$

so

$$f'(x) = \begin{cases} -1, & x < 2 \\ 1, & x > 2 \end{cases} = \begin{cases} 1, & x > 2 \\ -1, & x < 2 \end{cases}$$

If $x=2$

$$\lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h} = 1$$

$$\lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{h} = -1$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \text{ DNE}$$

if $x=2$

$$(b) \frac{f(x+h) - f(x)}{h} = \frac{\sqrt{x+h+3} - \sqrt{x+3}}{h} = \frac{1}{\sqrt{x+h+3} + \sqrt{x+3}}$$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \frac{1}{2\sqrt{x+3}} = f'(x)$$

f' DNE for $x = -3$

(f is not continuous at $x = -3$, only right continuous)

$$(c) \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x+1) = 2 \neq 4 = \lim_{x \rightarrow 1^+} f(x)$$

f is discontinuous at $x=1 \Rightarrow f$ is not differentiable at $x=1$